

LAKE WATEREE, SC, USA

WMO: 998208

Lat: 34.335N

Lon: 80.702W

Elev: 285

StdP: 14.54

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.0	(c) 25.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 18.9	(k) 44.7	(l) 17.3	(m) 45.2	(n) 2.9	(o) 140	(p) 0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.3	(c) 98.2	(d) N/A	(e) 95.7	(f) N/A	(g) 93.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.8	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.2	(b) 12.0	(c) 10.1	(e) 15.4	(f) 101.6	(g) 4.7	(h) 3.0	(i) 12.0	(j) 103.8	(k) 9.2	(l) 105.6	(m) 6.6	(n) 107.3	(o) 3.1	(p) 109.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	43.5	47.7	53.5	63.9	72.1	79.5	81.6	80.1	75.5	63.9	52.2	49.1		
	DBStd	15.08	9.15	9.03	8.53	6.65	5.80	4.34	3.53	3.71	5.18	6.95	7.25	8.79		
	HDD50	635	247	142	58	2	0	0	0	0	0	2	58	126		
	HDD65	2629	668	490	368	99	15	0	0	0	2	102	388	497		
	CDD50	5615	45	76	168	419	686	885	980	935	765	434	123	99		
	CDD65	2131	1	4	13	65	235	435	515	470	317	69	3	4		
	CDH74	21015	6	31	162	751	2307	4547	5324	4417	2770	667	25	8		
	CDH80	9851	0	1	27	182	946	2371	2729	2164	1252	179	0	0		
(14) Wind	WSAvg	3.5	4.0	4.1	4.3	4.1	3.3	3.1	3.1	2.6	2.9	3.5	3.6	3.6		
(15) Precipitation	PrecAvg	45.8	4.4	3.6	4.3	2.9	3.0	4.4	5.0	4.8	3.7	3.1	3.0	3.2		
	PrecMax	61.6	9.0	7.5	10.2	6.6	7.2	10.3	9.9	10.8	8.8	16.9	8.5	8.0		
	PrecMin	30.6	0.7	0.8	0.4	0.2	0.6	0.6	0.4	0.6	0.2	0.0	0.3	0.4		
	PrecStd	7.3	2.0	1.8	2.0	1.7	1.7	2.6	2.8	2.4	2.2	3.1	1.8	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.2	78.2	83.6	88.1	96.5	101.6	101.6	99.2	96.8	92.2	77.3	75.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	67.6	73.1	78.5	84.8	92.6	97.5	98.4	95.7	93.6	84.2	73.5	71.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	64.3	68.8	74.6	82.0	88.8	94.4	95.5	93.3	90.8	80.8	70.0	68.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	59.7	64.3	70.1	78.6	85.6	91.7	92.5	90.6	87.7	77.4	66.6	64.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	22.7	23.7	24.9	22.9	22.7	21.3	20.4	21.4	23.6	24.7	21.4		
		MCDBR	27.1	27.2	29.7	29.4	27.6	27.0	25.9	24.5	26.9	28.4	29.5	22.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.331	0.368	0.414	0.475	0.504	0.555	0.526	0.443	0.397	0.346	0.318		
	taud		2.495	2.463	2.389	2.248	2.128	2.088	1.968	2.039	2.255	2.352	2.457	2.539		
	Ebn at Noon		282	291	287	277	260	252	238	243	259	263	270	277		
	Edh at Noon		27	31	36	43	49	52	58	53	41	34	28	24		
(44) All-Sky Solar Radiation	RadAvg		869	1070	1414	1774	1944	2036	1924	1752	1507	1251	954	756		
	RadStd		45	115	105	133	157	143	119	83	128	151	76	69		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+324	+234

Nomenclature: See separate page